DAY 04 • STUDENT HANDOUT

Security Solutions, Detection Sources and Log Management

Where alerts come from, how security tools classify severity, and how logs are collected and read

Qualification	Cyber Threat Monitoring Level I
Unit of Competency	CS-ICT251101 Monitor and Report Cyber Threats
Coverage	Range of Variables: Detection alert sources and Security solutions; Required Knowledge 1.5 Sources of detection alerts, 1.6 Log and detection management, 1.8 Security solution management application, 1.9 Severity classifications
Duration	8 hours (1 training day)

Learning Objectives

At the end of this session, you should be able to:

1. Describe the purpose of each security solution listed in the Competency Standards (AV, firewall, WAF, IDS/IPS, DLP, EDR/MDR, NDR, SIEM, SOAR, XDR, threat intelligence, OT and IoT security).
2. Identify the source of a detection alert from its content.
3. Explain how logs are generated, collected, normalized and stored in a SIEM.
4. Interpret severity classifications used by security solutions.
5. Read firewall and Windows security logs to find suspicious patterns.

1. Layers of Defense

Organizations use **defense in depth** — several security layers so that if one fails, another detects or stops the threat. Each layer is a potential **source of detection alerts** that you will monitor.

Security Solution	What It Protects / Does	Typical Alert
Anti-virus (AV)	Scans files on endpoints for known malware signatures and heuristics	"Trojan:Win32/Agent detected — quarantined"
Common firewall	Allows/blocks traffic by IP, port, protocol	"Denied inbound TCP 3389 from 192.0.2.8"
NGFW	Next-generation firewall: firewall + application awareness + IPS	"Blocked: known C2 application traffic"
WAF	Web Application Firewall — protects websites from SQL injection, XSS, etc.	"SQL injection attempt blocked on /login.php"
IDS	Intrusion Detection System — watches traffic and alerts (does not block)	"ET SCAN Nmap scan detected"
IPS	Intrusion Prevention System — detects and blocks in-line	"Exploit attempt dropped"
DLP	Data Loss Prevention — stops sensitive data leaving the organization	"File with 200 credit card numbers sent to Gmail"

Security Solution	What It Protects / Does	Typical Alert
EDR	Endpoint Detection and Response — records endpoint behavior and can isolate hosts	"Suspicious PowerShell spawned by Word"
MDR	Managed Detection and Response — EDR/XDR operated by an outside provider	Provider escalation email/call
NDR	Network Detection and Response — analyzes network traffic behavior	"Beaconing to rare external domain"
TDR	Threat Detection and Response — platform detecting and responding across sources	Correlated threat case
SIEM	Security Information and Event Management — collects and correlates logs from everything	"Brute force followed by successful login"
SOAR	Security Orchestration, Automation and Response — automates playbooks	"Playbook: phishing — URL blocked, ticket created"
XDR	Extended Detection and Response — combines endpoint, network, email, cloud data	Multi-stage incident
Threat intelligence	Feeds of known-bad IPs, domains, hashes, actor information	"Connection to IP on threat list"
OT/ICS/SCADA security	Protects industrial systems (power, water, manufacturing)	"Unauthorized PLC write command"
IoT security	Protects smart devices (CCTV, sensors, smart TVs)	"CCTV camera contacting unknown server"

> **NOTE — Detection alerts in the Competency Standards**
>
> The CS lists these alert sources: **SIEM alert, AV alert, Firewall alert, Web application firewall alert, DLP alert, EDR alert and NDR alert.**

2. Log and Detection Management

A **log** is a time-stamped record of an event. A **detection** is a log or event that a security tool considers suspicious. Log management is the process of collecting, storing, protecting and reviewing logs.

The Log Pipeline

1. **Generate** — devices (servers, endpoints, firewalls, applications) produce logs.
2. **Collect** — logs are sent by agents (e.g., Wazuh agent, Winlogbeat) or via syslog to a collector.
3. **Normalize** — different formats are converted into common fields (time, source IP, user, action).
4. **Correlate** — the SIEM applies rules that connect events (e.g., 20 failed logins + 1 success = alert).
5. **Alert** — the SIEM creates an alert with a severity and sends it to the SOC queue.
6. **Store and retain** — logs are kept for investigations and compliance (often 90 days online, 1 year archived).

Key Log Fields

Field	Why It Matters
Timestamp + time zone	Builds the timeline; confirm the time zone (Philippines = UTC+8)
Host / device name	Identifies the affected asset
Source and destination IP/port	Shows who talked to whom; distinguishes ingress vs egress
User account	Shows who did it — or whose account was misused
Process / file / hash	Identifies the program involved
Action / result	Allowed, blocked, quarantined, failed
Rule / signature name and severity	Why the tool raised the alert and how serious it thinks it is

Sample Firewall Log Line

```
2026-11-04 10:15:22 FW-01 action=DENY proto=TCP src=192.0.2.8:51 234 dst=203.0.113.10:3389 rule=Block-RDP-Inbound
```

3. Security Solution Management Applications

Each security solution has a **management console** where analysts see alerts, endpoint status, and actions. Examples: Microsoft Defender portal, Wazuh dashboard, firewall web GUI, and SIEM search interfaces. In each console, learn where to find: the **alert list**, **device/agent status**, **detection details**, **actions taken**, and **logs**.

4. Severity Classifications

Security solutions assign a **severity** to each alert. Vendors use different scales, so SOPs map them to a common scale.

Common Scale	Typical Vendor Values	Meaning	Example
Critical	Critical / 13–15 (Wazuh) / P1	Active, high-impact threat; act immediately	Ransomware encrypting files
High	High / 10–12 / P2	Likely compromise needing urgent action	Malware executing, not yet contained
Medium	Medium / 7–9 / P3	Suspicious activity needing review	Port scan from the internet
Low	Low / 4–6 / P4	Minor or blocked activity	Adware blocked by AV
Informational	Info / 0–3	Normal events for records	Successful signature update

> **TIP**
> A tool's severity is a **starting point, not the final answer**. A "Low" alert on the domain controller may matter more than a "Medium" alert on a test PC. You will learn to assess alerts against pre-identified criteria on Day 6.

Key Terms

Term	Meaning
Defense in depth	Using multiple security layers so one failure does not expose everything
SIEM	Platform that collects, normalizes and correlates logs to generate alerts
EDR / XDR	Endpoint-focused / extended (multi-source) detection and response
WAF	Firewall that inspects web application requests
Normalization	Converting different log formats into common fields
Correlation rule	SIEM logic that links several events into one alert

References

- TESDA Competency Standards — Cyber Threat Monitoring Level I (ICT Sector), units CS-ICT251101 and CS-ICT251102.
- NIST SP 800-115, Technical Guide to Information Security Testing and Assessment (Chapter 4).
- NIST SP 800-61 Rev. 2/Rev. 3, Computer Security Incident Handling Guide.
- MITRE ATT&CK® Framework — attack.mitre.org; Lockheed Martin Cyber Kill Chain® .
- Republic Act No. 10175 (Cybercrime Prevention Act of 2012) and Republic Act No. 10173 (Data Privacy Act of 2012).